YOLO

Soma Biswas
Department of Electrical Engineering,
Indian Institute of Science, Bangalore.

Slide Credit:LShivang Singh, UT Austin

YOLO - Main features

1. Extremely fast (45 frames per second)

2. Reason Globally on the Entire Image

3. Learn Generalizable Representations

Key Insights

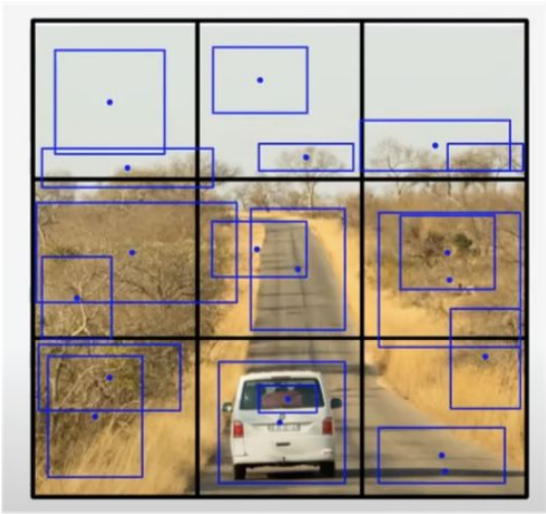
Previous Approaches

- A separate model for generating bounding boxes and for classification (more complicated model pipeline)
- Need to run classification many times (expensive computation)
- Looks at limited part of the image (lacks contextual information for detection)

YOLO algorithm

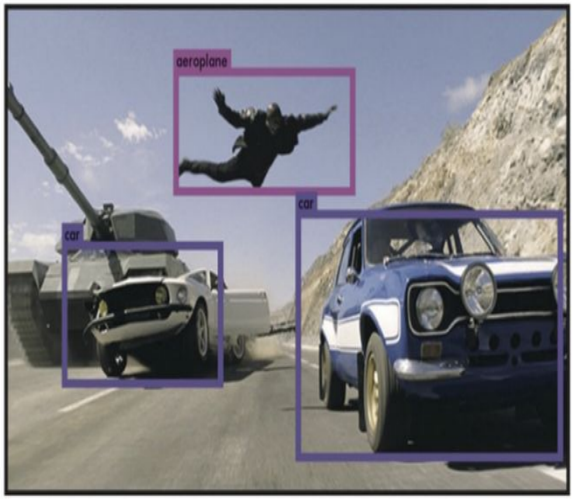
- A single neural network for localization and for classification (less complicated pipeline)
- Need to inference only once (efficient computation)
- Looks at the entire image each time leading to less false positives (has contextual information for detection)

Problem Setting



S = 3, B = 2

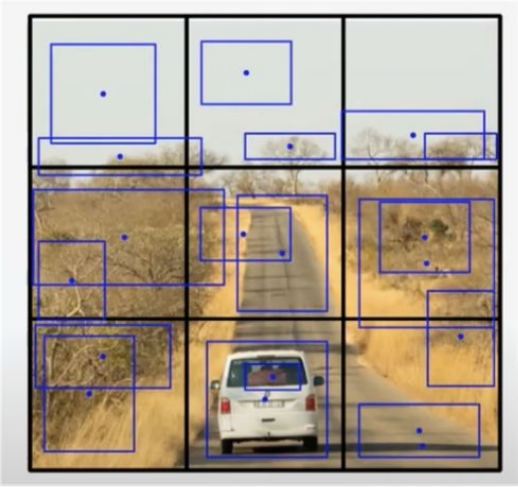
- Given an image generate bounding boxes, one for each detectable object in image
- For each bounding box, output 5 predictions: x, y, w, h, confidence. Also output class



- x, y (coordinates for center of bounding box)
- w,h (width and height)
- confidence (probability bounding box has object)
- class (classification of object in bounding box)

YOLO Overview

- First, image is split into a SxS grid
- For each grid square, generate B bounding boxes
- For each bounding box, there are 5 predictions: x, y, w, h, confidence



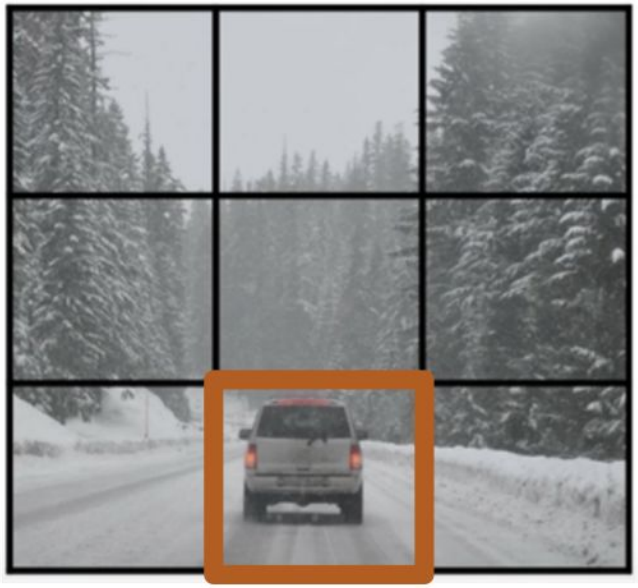
S = 3, B = 2

YOLO Training

For each grid block, we have a vector like this. For this example B is 2 and C is 2

ρ_1
b_x_1
b_y_1
b_h_1
b_w_1
ρ_2
b_x_2
b_y_2
b_h_2
b_w_2
c_1
c_2

- YOLO is a regression algorithm. What is X? What is Y?
- X is simple, just an image width (in pixels) * height (in pixels) * RGB values
- Y is a tensor of size S * S * (B * 5 + C)
- B*5 + C term represents the predictions + class predicted distribution for a grid block



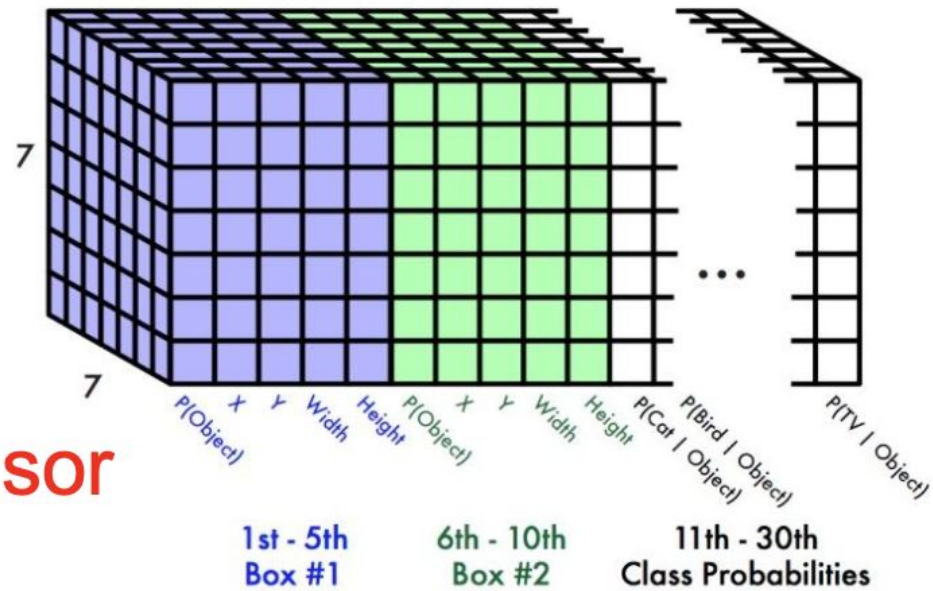
GT label example:

1
b_x_1
b_y_1
b_h_1
b_w_1
0
?
?
?
?
c_1 = 1
c_2 = 0

Final Output

Each cell predicts:

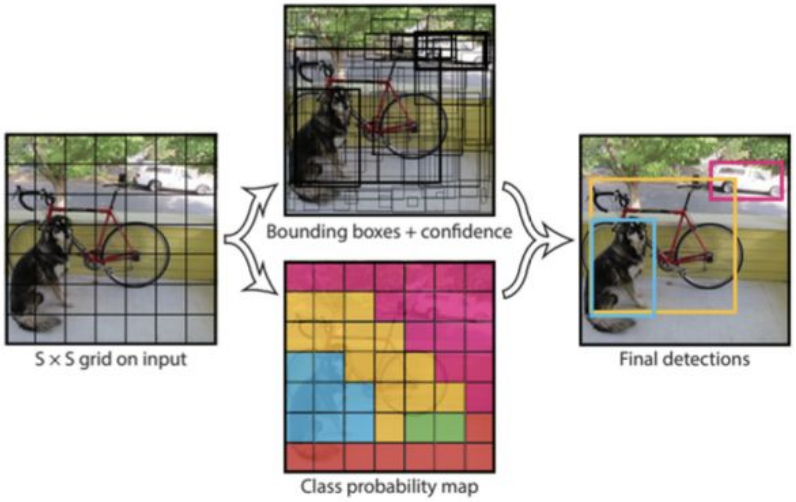
- For each bounding box:
  - 4 coordinates (x, y, w, h)
  - 1 confidence value
- Some number of class probabilities



S * S * (B * 5 + C) tensor

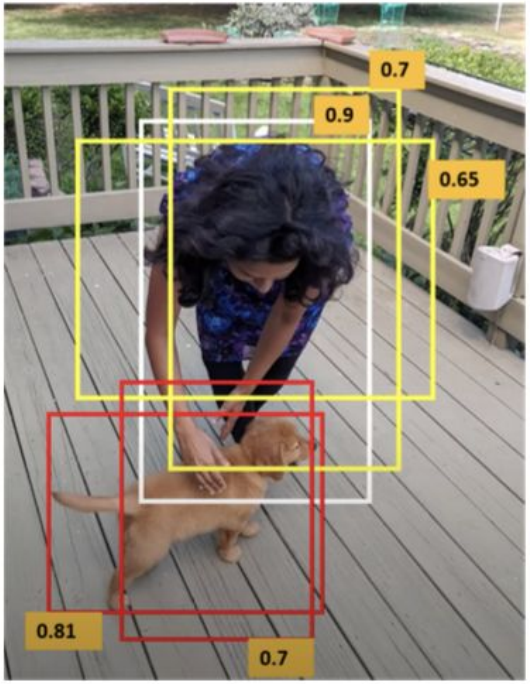
YOLO Prediction

- We then use the output to make final detections
- Use a threshold to filter out bounding boxes with low P(Object)
- In order to know the class for the bounding box compute score take argmax over the distribution Pr(Class|Object) for the grid the bounding box's center is in



NMS: Non-maxima suppression

- Most of the time objects fall in one grid, however it is still possible to get redundant boxes (rare case as object must be close to multiple grid cells for this to happen)
- Discard bounding box with high overlap (keeping the bounding box with highest confidence)
- Adds 2-3% on final mAP score



YOLO Training Objective

- For YOLO, we need to minimize the following loss
- Sum squared error is used

$$\lambda_{\text{coord}} \sum_{i=0}^{S^2} \sum_{j=0}^{B} \mathbb{1}_{ij}^{\text{obj}} \left[(x_i - \hat{x}_i)^2 + (y_i - \hat{y}_i)^2 \right]$$
$$+ \lambda_{\text{coord}} \sum_{i=0}^{S^2} \sum_{j=0}^{B} \mathbb{1}_{ij}^{\text{obj}} \left[\left(\sqrt{w_i} - \sqrt{\hat{w}_i} \right)^2 + \left(\sqrt{h_i} - \sqrt{\hat{h}_i} \right)^2 \right]$$

Coordinate Loss: Minimize the difference between x,y,w,h pred and x,y,w,h ground truth. ONLY IF object exists in grid box and if bounding box is resp for pred

$$+ \sum_{i=0}^{S^2} \sum_{j=0}^{B} \mathbb{1}_{ij}^{\text{obj}} \left(C_i - \hat{C}_i \right)^2$$

Confidence Loss: Loss based on confidence ONLY IF there is object

$$+ \lambda_{\text{noobj}} \sum_{i=0}^{S^2} \sum_{j=0}^{B} \mathbb{1}_{ij}^{\text{noobj}} \left(C_i - \hat{C}_i \right)^2$$

No Object Loss based on confidence if there is no object

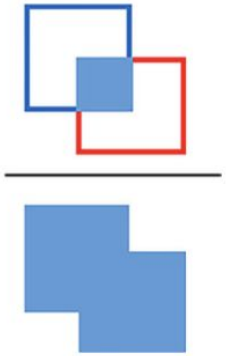
$$+ \sum_{i=0}^{S^2} \mathbb{1}_{i}^{\text{obj}} \sum_{c \in \text{classes}} (p_i(c) - \hat{p}_i(c))^2$$

Class loss, minimize loss between true class of object in grid box

Experiments

- Authors compare YOLO against the previous work described above on PASCAL VOC 2007, and VOC 2012 as well as out of domain art dataset
- Correct if IOU metric above .5 and class is correct
- Use two performance metrics:
  - mAP score: mean average precision
  - FPS: frames per second
- Add FAST YOLO: which has less parameters

$$IoU = \frac{area\ of\ intersection}{are\ of\ union} =$$

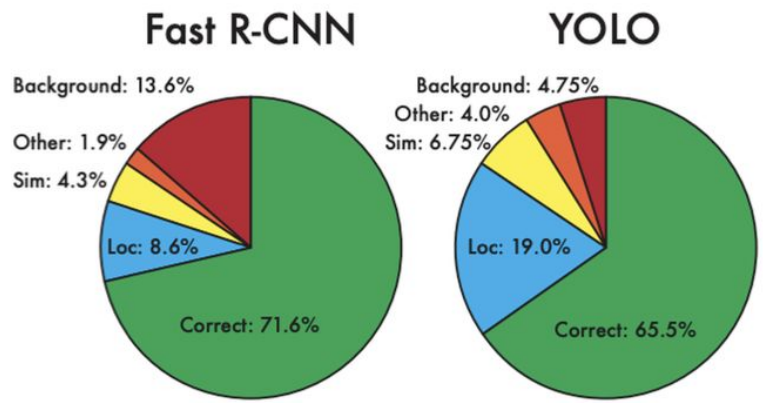


- Baseline YOLO outperform real time detectors by large amount
- Do better than most less than real time as well

Real-Time Detectors	Train	mAP	FPS
100Hz DPM [31]	2007	16.0	100
30Hz DPM [31]	2007	26.1	30
Fast YOLO	2007+2012	52.7	**155**
YOLO	2007+2012	**63.4**	45
Less Than Real-Time			
Fastest DPM [38]	2007	30.4	15
R-CNN Minus R [20]	2007	53.5	6
Fast R-CNN [14]	2007+2012	70.0	0.5
Faster R-CNN VGG-16[28]	2007+2012	73.2	7
Faster R-CNN ZF [28]	2007+2012	62.1	18
YOLO VGG-16	2007+2012	66.4	21

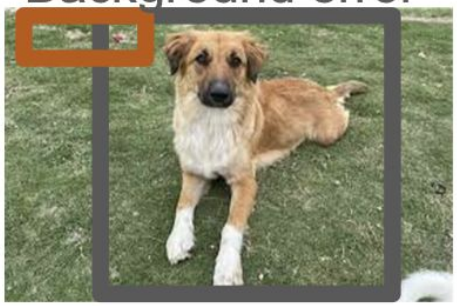
Error Analysis

- Makes far less background errors (less likely to predict false positives on background)
  - IOU is VERY small with any ground truth label
- But far more localization errors
  - Correct class, IOU is somewhat small



Localization error

Background error



Out-of-Domain Results

- Ran YOLO + competitors (trained on natural images) on art
- Does well on artistic datasets where more having global context greatly helps

	VOC 2007	Picasso		People-Art
	AP	AP	Best F_1	AP
YOLO	**59.2**	**53.3**	**0.590**	**45**
R-CNN	54.2	10.4	0.226	26
DPM	43.2	37.8	0.458	32
Poselets [2]	36.5	17.8	0.271	
D&T [4]	-	1.9	0.051	

Discussion

- Pro: YOLO is a lot faster than the other algorithms for image detection
- Pro: YOLO's use of global information rather than only local information allows it to understand contextual information when doing object detection
  - Does better in domains such as artwork due to this
- Con: YOLO lagged behind the SOTA models in object detection
  - This is attributed to making many localization errors and unable to detect small object